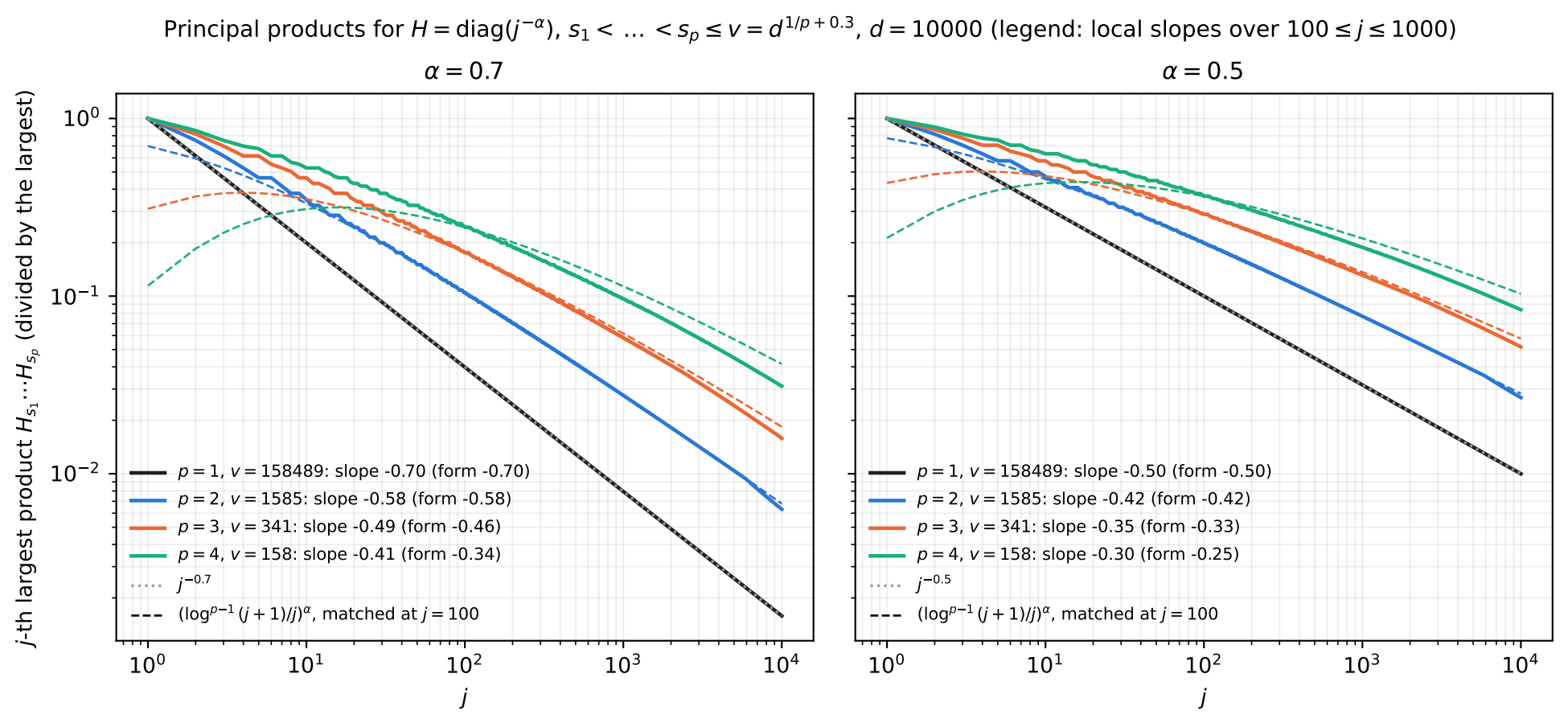

Principal products for $H = \mathrm{diag}(j^{-\alpha})$, $s_1 < \ldots < s_p \le v = d^{1/p+0.3}$, $d = 10000$ (legend: local slopes over $100 \le j \le 1000$)
$\alpha = 0.7$
$\alpha = 0.5$
j-th largest product $H_{s_1} \cdots H_{s_p}$ (divided by the largest)
j
$p = 1$, $v = 158489$: slope -0.70 (form -0.70)
$p = 2$, $v = 1585$: slope -0.58 (form -0.58)
$p = 3$, $v = 341$: slope -0.49 (form -0.46)
$p = 4$, $v = 158$: slope -0.41 (form -0.34)
$j^{-0.7}$
$(\log^{p-1}(j+1)/j)^{\alpha}$, matched at $j = 100$
$p = 1$, $v = 158489$: slope -0.50 (form -0.50)
$p = 2$, $v = 1585$: slope -0.42 (form -0.42)
$p = 3$, $v = 341$: slope -0.35 (form -0.33)
$p = 4$, $v = 158$: slope -0.30 (form -0.25)
$j^{-0.5}$
$(\log^{p-1}(j+1)/j)^{\alpha}$, matched at $j = 100$